

Pandas for Data *Analysis*

Work your own messy dataset through orient, select, clean, and aggregate — leave with real numbers, not just notes.

NAME

DATE

DATASET USED

1 Step 1 — Orient: Your First 60 Seconds with the Dataset

- Name your dataset — **file** or **description**:
- Run `df.shape`, `df.info()`, `df.head()`. Rows / columns:
- List columns with **null values**:
- Which *dtypes* look wrong (dates as object, numbers as string)?
- First instinct — what's this data trying to tell you?

SHAPE & NULLS

2 Step 2 — Select: Define the Question Before You Filter

- Write the **ONE question** this dataset must answer:
- List the columns *essential* to answering it:
- Write your `.loc` or boolean filter that isolates the rows you care about:
- What rows get **excluded** (test data, outliers, nulls) — why?
- Rows remaining after filter — does that count feel right?

FILTER LOGIC

3 Step 3 — Clean: Audit the Mess Before You Trust the Numbers

- List columns needing **type conversion** and target type:
- How is missing data encoded? `NaN`, `'N/A'`, `'-'`, blank — list every variant found:
- Duplicate rows — run `df.duplicated().sum()`. Count:
- Which *string columns* have inconsistent casing/whitespace?
- Name the **one cleaning step** that most improves this dataset:

CLEANING CHECKLIST

4 Step 4 — Aggregate: Find the Summary That Actually Tells You Something

- Groupby column — **region**, **category**, **date**, **segment**?
- Aggregation that fits — *sum*, *mean*, *count*, *median*? Why?
- Run the groupby. Highest group / lowest group:
- Is one group carrying a **disproportionate share** (>30–40%)?
- What follow-up question does this raise the raw data couldn't answer?

GROUPBY RESULTS

5 Reflection: What the Data Actually Said

- Most **surprising** find during cleaning — not analysis:
- Did your hypothesis hold after groupby — or did the summary *contradict* it?
- What breaks if you skip cleaning and go straight to aggregation?
- The **one insight** you'd hand a non-technical stakeholder:
- Next Pandas operation you need to go deeper on this dataset:

TAKEAWAYS